

Sample End Semester Question Paper - 2025

Subject: Engineering Mechanics | Code: ME202 | Semester: 2

Branch: Mechanical | Regulation: R20 (sample) | Time: 3 Hours | Max I

This is a SAMPLE question paper generated for Backlog Buddy (demo/testing data).
It is not an official university paper and is provided to demonstrate the platform.

PART A - Short questions (10 x 1 = 10 Marks)

1. Define angle of repose.
2. What is a truss?
3. State Varignon's theorem.
4. What is the perpendicular axis theorem?
5. Define resultant force.
6. State Lami's theorem.
7. What is the moment of a force?
8. Define centroid.
9. State the parallel axis theorem.
10. What is friction?

PART B - Long questions (6 x 13 = 78 Marks)

1. Find the resultant of a system of concurrent forces with a diagram.
2. Explain and apply Lami's theorem and Varignon's theorem.
3. Find the centroid of a given composite section.
4. Explain moment of inertia and the parallel/perpendicular axis theorems with examples.
5. Solve a friction problem: wedge and angle of repose.
6. Analyze a truss using method of joints and method of sections.